



Elier Garcia

ELECTRICAL ENGINEER · INDUSTRIAL
INSTALLATIONS

Contact

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Skills

Electrical Engineering

AutoCAD · OPUS · LV/MV Installations ·
NEC/IEC/NOM Standards · Site
supervision · Subcontractor management
· Office 365

Backend & Cloud

Supabase · PostgreSQL · Docker (Self-
hosted)

AI / ML & Data

Gemini AI · RAG Architectures · pgvector
· Python · SQL

DevOps & Tooling

n8n · Power BI

Languages

Spanish (native)
English B2
German (basic)

Awards & certifications

Electric Power Systems — University at
Buffalo — SUNY · February 2023

PVC-Coated Conduit Installer — Plasti-
Bond · April 2024

**Artificial Intelligence — Principles &
Applications** — Universidad del Valle de
México · August 2026

Digital Leadership Programme —
INESDI Business TechSchool · April 2025

Profile

Electrical engineer with two years at DEMEK on large-scale industrial projects: one year estimating electrical installations and one on site, first at Microsoft's hyperscale data center in Querétaro, where I went from newest resident to sole lead of the outdoor front, and then at BMW Group's plant in San Luis Potosí. M.S. in Business Analytics & AI with the top record of my cohort: I apply data and automation to the problems I lived on site, such as materials tracking and technical documentation.

Work experience

Founder · AI & Automation Engineer

2026 — present

EG Solutions · Madrid (remote)

- Design and development of web platforms, n8n automations and AI agents for businesses.

Electrical Construction Resident

Jan 2024 — Feb 2025

DEMEK S.A. de C.V. · Querétaro / San Luis Potosí, Mexico

- **Microsoft Azure hyperscale Data Center** (Querétaro, 9 months): from newest resident to **sole lead of the outdoor electrical front** in 6 months, directly supervising **60+ people**, with daily progress meetings, weekly planning and coordination with mechanical, telecom and civil works.
- Outdoor scope: medium voltage (conduit and cabling, coordinated with civil works), grounding systems, installation of transformers, generators and switchgear with crane coordination, pump room (MCC, lighting and power panels), outdoor pole lighting and temporary site power.
- Procurement of critical long-lead materials through requisitions, planning and follow-up with suppliers.
- **BMW Group plant** (San Luis Potosí, 5 months, 19,847 m² body-shop expansion in 8 sections): supervised LV and grounding in 2 sections (~40 people) and handled material requisitions for nearly all 8.
- Built my own materials tracking system (Excel + SAP): every off-budget request technically justified (take-off vs. drawings or documented design change), fewer supply stoppages, and irregularities detected that led to safety improvements.
- Compliance with NEC/IEC/NOM specifications and safety standards on both sites.

Electrical Budget Analyst

Jan 2023 — Jan 2024

DEMEK S.A. de C.V. · Chihuahua, Mexico

- Full estimating cycle for several projects in parallel (1–2 week deadlines): drawings and specs in AutoCAD, Excel quantity take-offs by room, system and area, and OPUS pricing by line item.
- Supplier quotes under client-defined brand and function specs, full indirect costs (security, site offices, cranes, labor, safety and documentation) and change orders during execution.
- Won the **Terex project in Nuevo León**: the only bid that reached the final presentation, delivered in English directly to Terex through Copachisa.

Related project

PDF Technical Assistant

pdf-tech-assistant2.vercel.app

Built from my own site experience to query electrical specs and technical manuals: each company uploads its PDFs and gets answers only from the documents, focused on its specialty (electrical, civil, mechanical...). Multi-company with roles, delivered as SaaS or on-premise with Docker.

React 18 · TypeScript · Supabase Edge Functions · pgvector · Gemini

Education

B.S. in Electrical Engineering

Instituto Tecnológico de Chihuahua · Mexico · 2023

Major in electrical installations. GPA 9.27/10. President of the Electrical Engineering student committee.

M.S. in Business Analytics & Artificial Intelligence

INESDI Business TechSchool · Madrid, Spain · 2026

Top academic record of the cohort (8.94/10). ETL/ELT, RAG architectures, applied machine learning and Power BI.